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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: DATA SCIENCE 24AD3107

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TITLE OF THE PROJECT	Similarity-Based Duplicate Record Detection System	
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Abstract

The **Similarity-Based Duplicate Record Detection System** is a Data Science project designed to identify exact and near-duplicate records in structured datasets using data preprocessing, data quality analysis, similarity and dissimilarity measures, Exploratory Data Analysis (EDA), and performance evaluation. The system begins by loading and inspecting datasets and handling missing values, noise, inconsistent formats, and existing duplicate entries. Records are standardized and prepared for comparison using appropriate measures such as Cosine Similarity, Jaccard Similarity, Levenshtein Distance, and Euclidean Distance, depending on whether attributes are textual, categorical, numerical, or set-based. Similarity scores and configurable thresholds are then used to classify records as duplicates, near-duplicates, or unique records. The project also performs EDA through summary statistics, distribution plots, box plots, correlation analysis, and heatmaps to understand dataset characteristics and duplicate patterns.

The system evaluates data quality based on completeness, consistency, uniqueness, and duplication, allowing comparison before and after the deduplication process. Where labeled duplicate records are available, performance is measured using Precision, Recall, F1-Score, Accuracy, and a Confusion Matrix. An interactive Python-based dashboard enables users to upload datasets, analyze data quality, detect duplicate records, compare similarity techniques, adjust thresholds, inspect matched record pairs and their similarity scores, visualize results, and download the cleaned dataset. By integrating preprocessing, similarity analysis, visualization, performance evaluation, and data quality assessment into a single workflow, the system provides an end-to-end approach for improving dataset reliability, reducing redundancy, and supporting accurate data analysis and machine learning applications across different domains.

Signature of the Guide

HOD